DBMS PBL · PROJECT 60

Disaster Relief Resource and Camp Management System

MySQL database + Python application

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25TU03MJM0 Database Management Systems

The problem

Which family still needs what?

Separate registers hide current needs and stock.

The system links families, assessed needs, donations and issues.

All example records in this project are fictional.

Three review stages

1

Design

Problem, requirements, ER diagram and table mapping

2

MySQL prototype

Schema, constraints, sample data, views and queries

3

Python application

Forms, reports, validation, tests and final report

How the database is organized

Camp **Family** **Person / Need**

One camp has many families. One family has many people and needs.

A distribution fulfills one assessed need and records who handled it.

PK + FK + UNIQUE + CHECK support data integrity.

Practical innovation using basic DBMS ideas

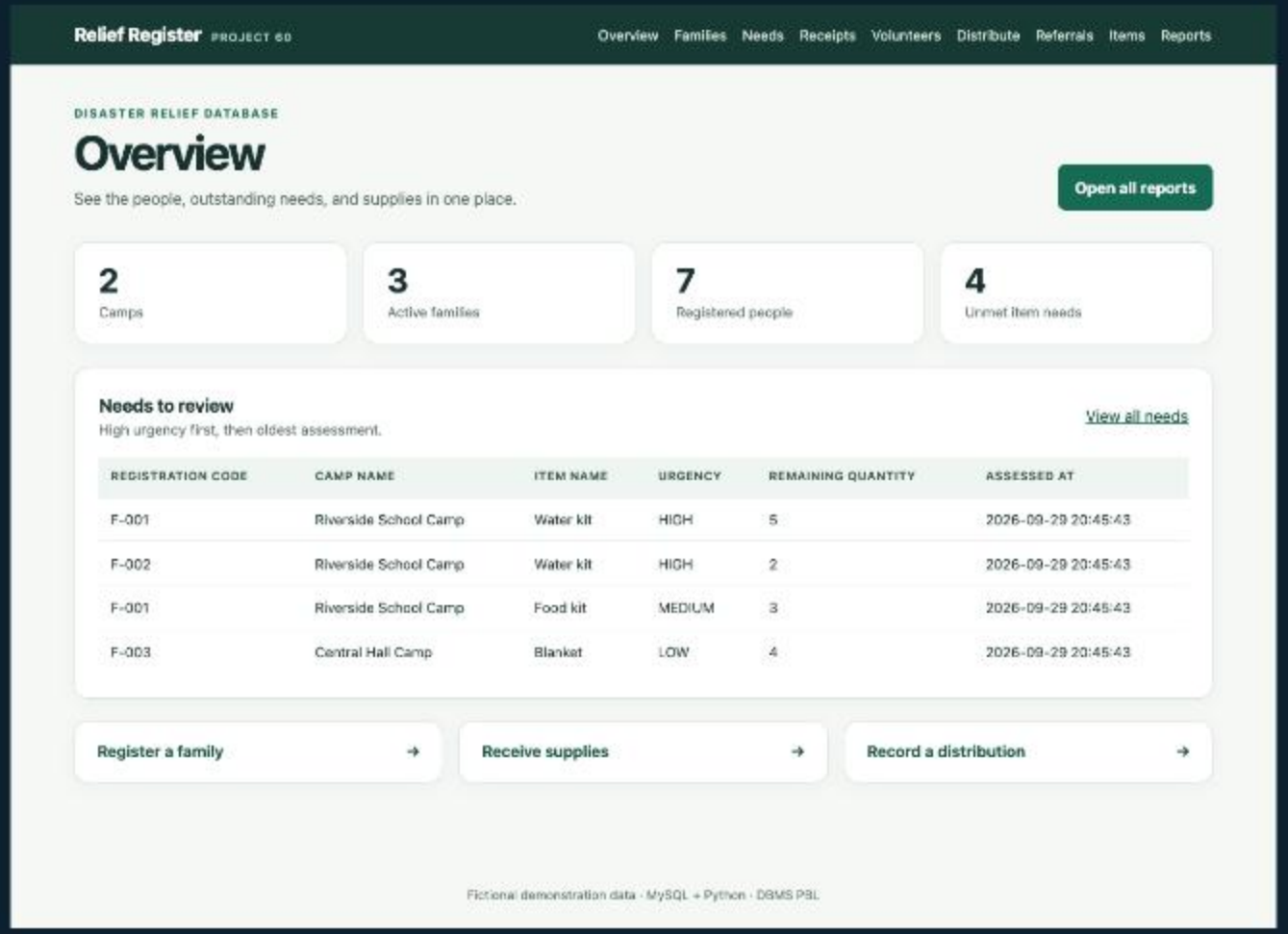
Remaining need = assessed quantity − issued quantity

Queue: HIGH, then MEDIUM, then LOW; oldest first within each level.

Issue transaction checks need, stock and volunteer assignment.

Receipt and issue movements explain each stock balance.

Recorded demo: unmet-needs queue



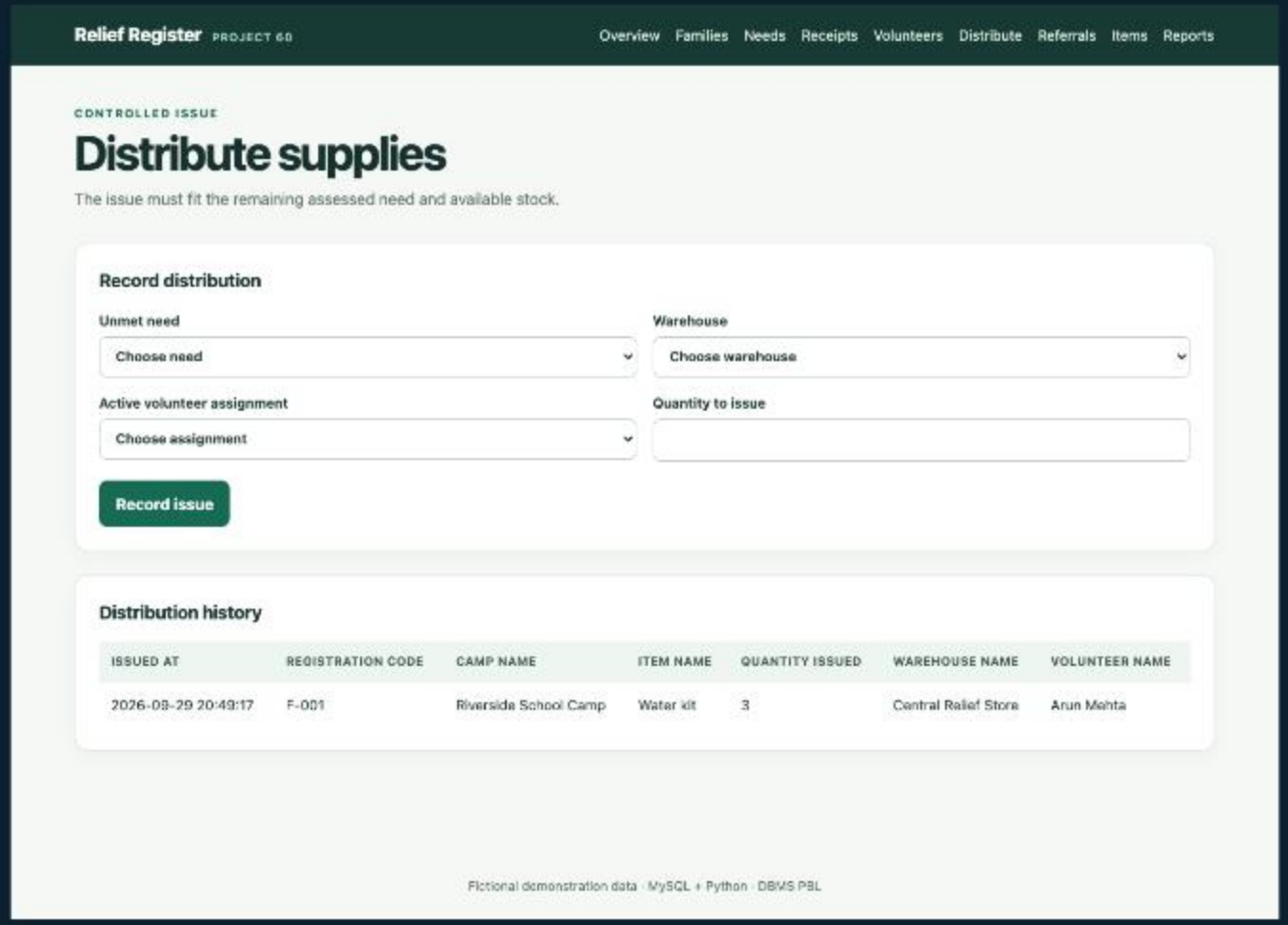
Overview

Camp and family totals

Unmet needs sorted by urgency

Open F-001 for the next step

Recorded demo: issue three water kits



F-001 water kits

Need: 5 to 2

Stock: 20 to 17

Excess issue is rejected

Verified reports after the issue

Unmet needs

REGISTRATION CODE	CAMP NAME	ITEM NAME	URGENCY	QUANTITY REQUIRED	QUANTITY ISSUED	REMAINING QUANTITY
F-001	Riverside School Camp	Water kit	HIGH	5	3	2
F-002	Riverside School Camp	Water kit	HIGH	2	0	2
F-001	Riverside School Camp	Food kit	MEDIUM	3	0	3
F-003	Central Hall Camp	Blanket	LOW	4	0	4

Warehouse stock

WAREHOUSE NAME	ITEM NAME	UNIT	QUANTITY ON HAND
Central Relief Store	Blanket	piece	10
Central Relief Store	Food kit	kit	12
Central Relief Store	Water kit	kit	17

F-001: 2 needed · Store: 17 water kits

What the project demonstrates

8 MySQL integration tests passed

Duplicate and excess issues are rejected.

Stock reconciles with the movement history.

The app uses fictional data on a local computer.